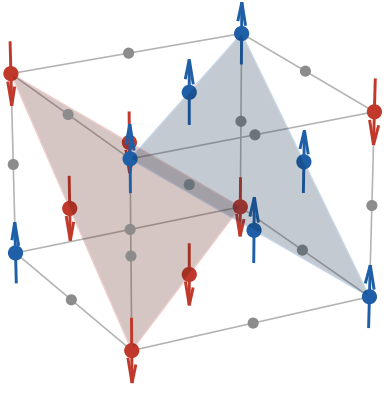
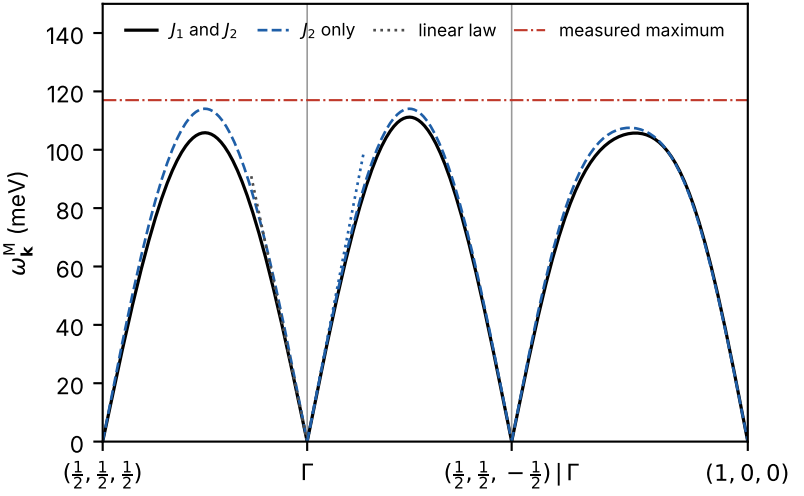
(a) NiO, type-II order



(b) linear spin waves, wavevectors in units of $2\pi/a$